

TCN2141 Computer Networks**Lab 1 (6 marks)**

Important instruction: Print lab sheet and submit it with all solution derived from lab experiment to tutors.

Lab 1**Part A: Understanding and Using Packet Tracer**

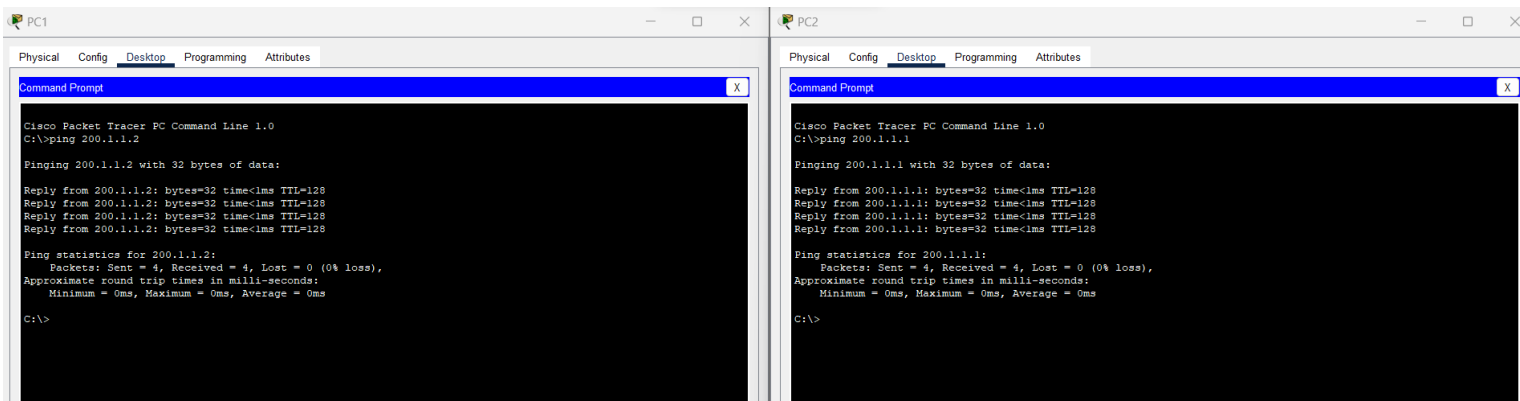
Use Cisco Packet Tracer

1.1 Draw and simulate the following networks. Note there are 3 labs: a, b and c.

a) Ensure PC1 can ping PC2 and vice-versa.

Network mask: 255.255.255.0

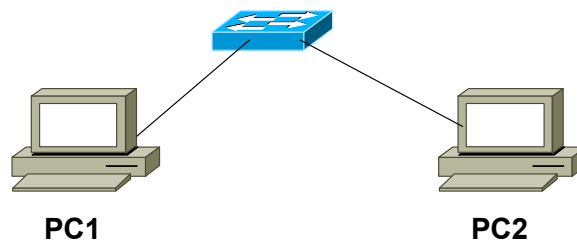
IP addresses: PC1: 200.1.1.1; PC2: 200.1.1.2

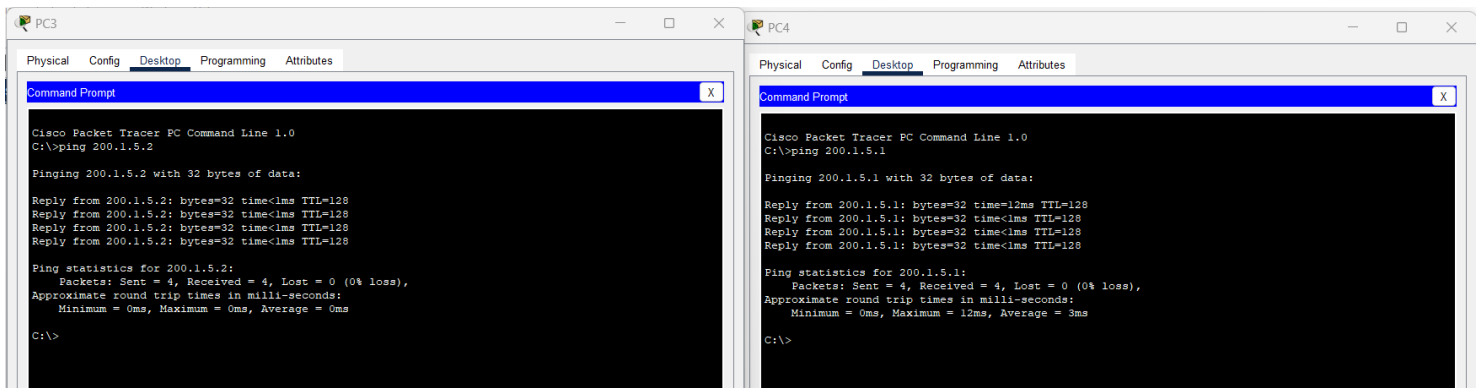


b) PC1 can ping PC2 and vice-versa.

Network mask: 255.255.255.0

IP addresses: PC1: 200.1.5.1; PC2: 200.1.5.2





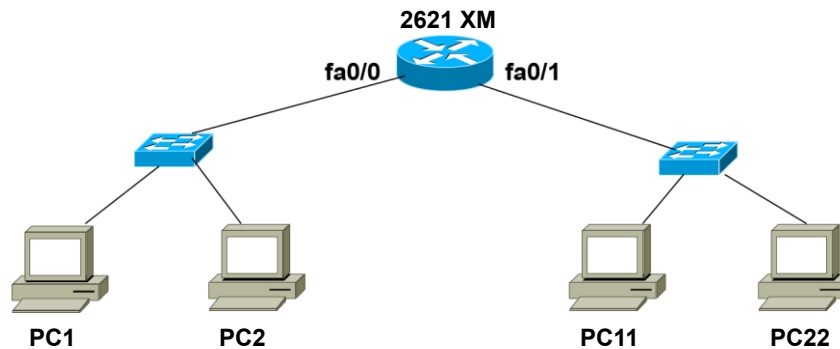
c) PC1, PC2, PC11 and PC22 can ping each other.

Network mask: 255.255.255.0

IP addresses: PC 1: 192.168.1.2; PC 2: 192.168.1.4; PC 11: 192.168.2.2; PC 22: 192.168.2.4

You must configure the fa0/0 to IP address: 192.168.1.254

You must configure the fa0/1 to IP address: 192.168.2.254



Part B: To configure the IP Address & routing on the router

Window-based Steps

- Double-click the router. A pop-up window will come up
- Select the middle sub-menu (config)
- To configure IP address, select the interfaces
- To configure routing, select routing

Command Line Steps: Note: The **Bold** character is what you need to type

Click the CLI to configure the IP Address on the router.

Step 1: Global Configuration Mode

Router>

Router>**enable**

Router# **configure terminal**

Step 2: Configuring a Router Name

Router (config)#**hostname Router 1**

Router1 (config)#

Step 3: Creating a MOTD Banner

Router1(config)#**banner motd # This is a secure system. Authorized Personnel Only! #**
Router1(config)#

Step 4: Configuring Passwords - Sets enable password and sets enable secret password

Router1(config)#**enable password cisco**
Router1(config)#**enable secret class**
Router1(config-line)#**exit**

Step 5: Configuring Passwords for Console port and Telnet

Router 1(config)#**line console 0**
Router 1(config-line)# **password cisco**
Router 1(config-line)# **login**

Router 1(config)#**line vty 0 4**
Router1(config-line)# **password cisco**
Router1(config-line)# **login**

Step 6: Configuring an Fast Ethernet Interface

Router 1(config)#**interface fa0/0**
Router 1(config-if)#**description LAN 1 interface**
Router 1(config-if)#**ip address 192.168.1.254 255.255.255.0**
Router 1(config-if)#**no shutdown**
Router 1(config-if)#**exit**

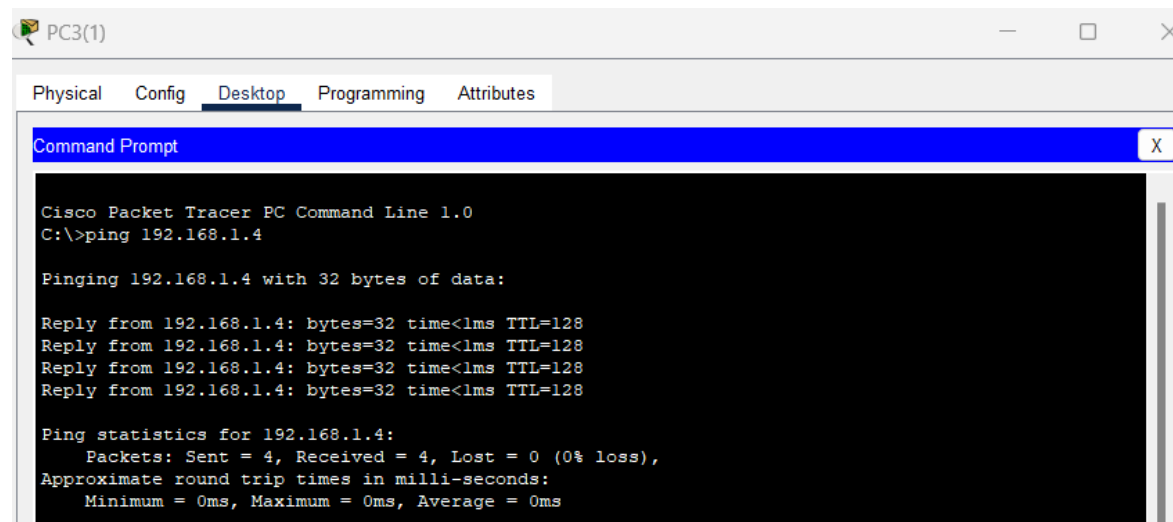
Router 1(config-if)#**interface fa0/1**
Router 1(config-if)#**description LAN 2 interface**
Router 1(config-if)#**ip address 192.168.2.254 255.255.255.0**
Router 1(config-if)#**no shutdown**
Router 1(config-if)#**exit**
Router1(config)#**exit**

Step 7: Saving Configurations

Route 1# **copy run start**

PC1 (PC3(1))

PC2



The screenshot shows a Cisco Packet Tracer interface for PC3(1). The 'Desktop' tab is active, displaying a 'Command Prompt' window. The window title is 'Command Prompt' with a close button. The text inside the window shows the execution of a ping command to 192.168.1.4. The output indicates that all four packets were received successfully with 0% loss.

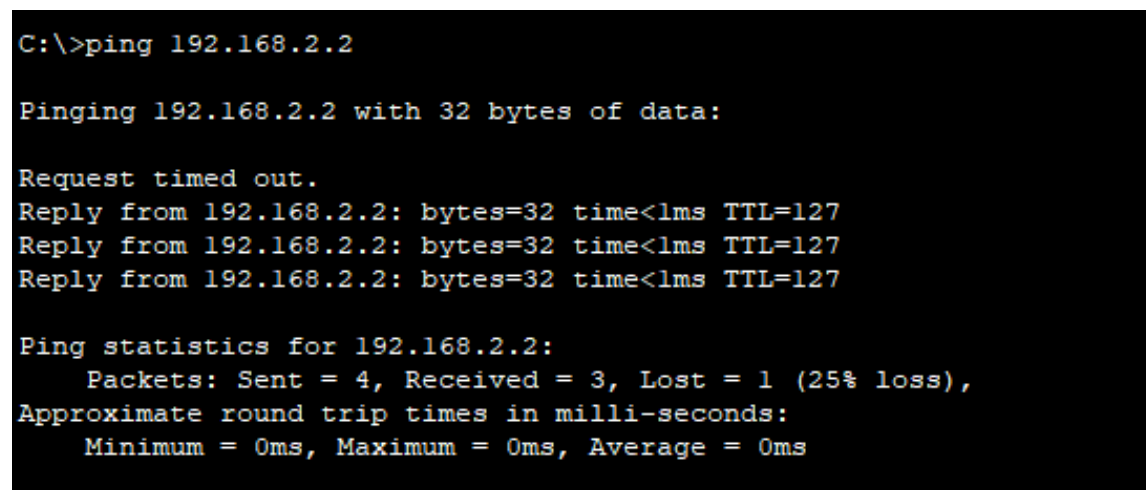
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC11



The screenshot shows the Command Prompt for PC11. The user has entered a ping command to 192.168.2.2. The output shows that one packet was lost, resulting in a 25% loss rate.

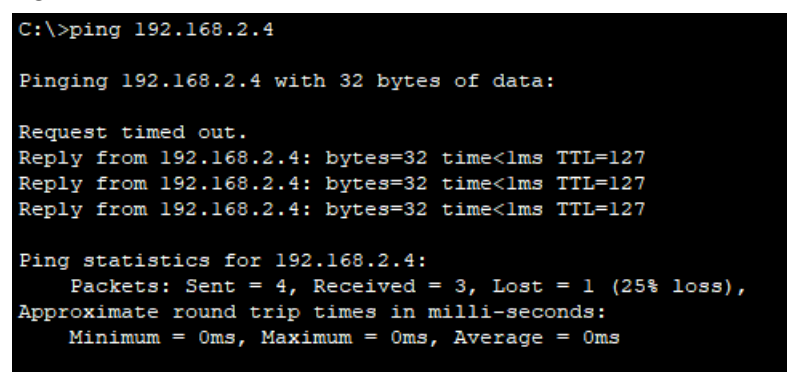
```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC22



The screenshot shows the Command Prompt for PC22. The user has entered a ping command to 192.168.2.4. The output shows that one packet was lost, resulting in a 25% loss rate.

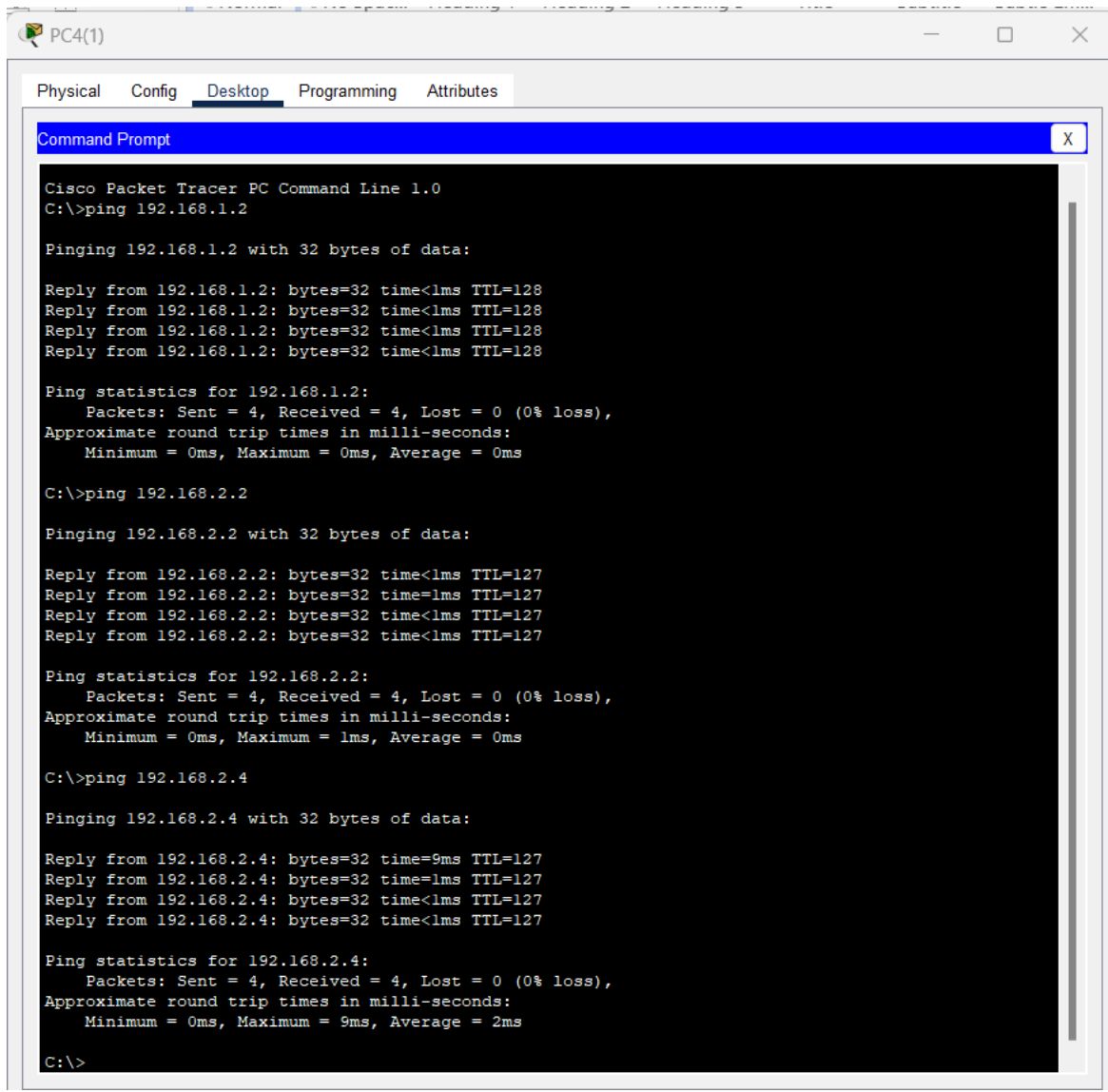
```
C:\>ping 192.168.2.4

Pinging 192.168.2.4 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.4: bytes=32 time<1ms TTL=127
Reply from 192.168.2.4: bytes=32 time<1ms TTL=127
Reply from 192.168.2.4: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC2 (PC4(1))



```
PC4(1)
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<lms TTL=128
Reply from 192.168.1.2: bytes=32 time<lms TTL=128
Reply from 192.168.1.2: bytes=32 time<lms TTL=128
Reply from 192.168.1.2: bytes=32 time<lms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<lms TTL=127
Reply from 192.168.2.2: bytes=32 time<lms TTL=127
Reply from 192.168.2.2: bytes=32 time<lms TTL=127
Reply from 192.168.2.2: bytes=32 time<lms TTL=127

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.2.4

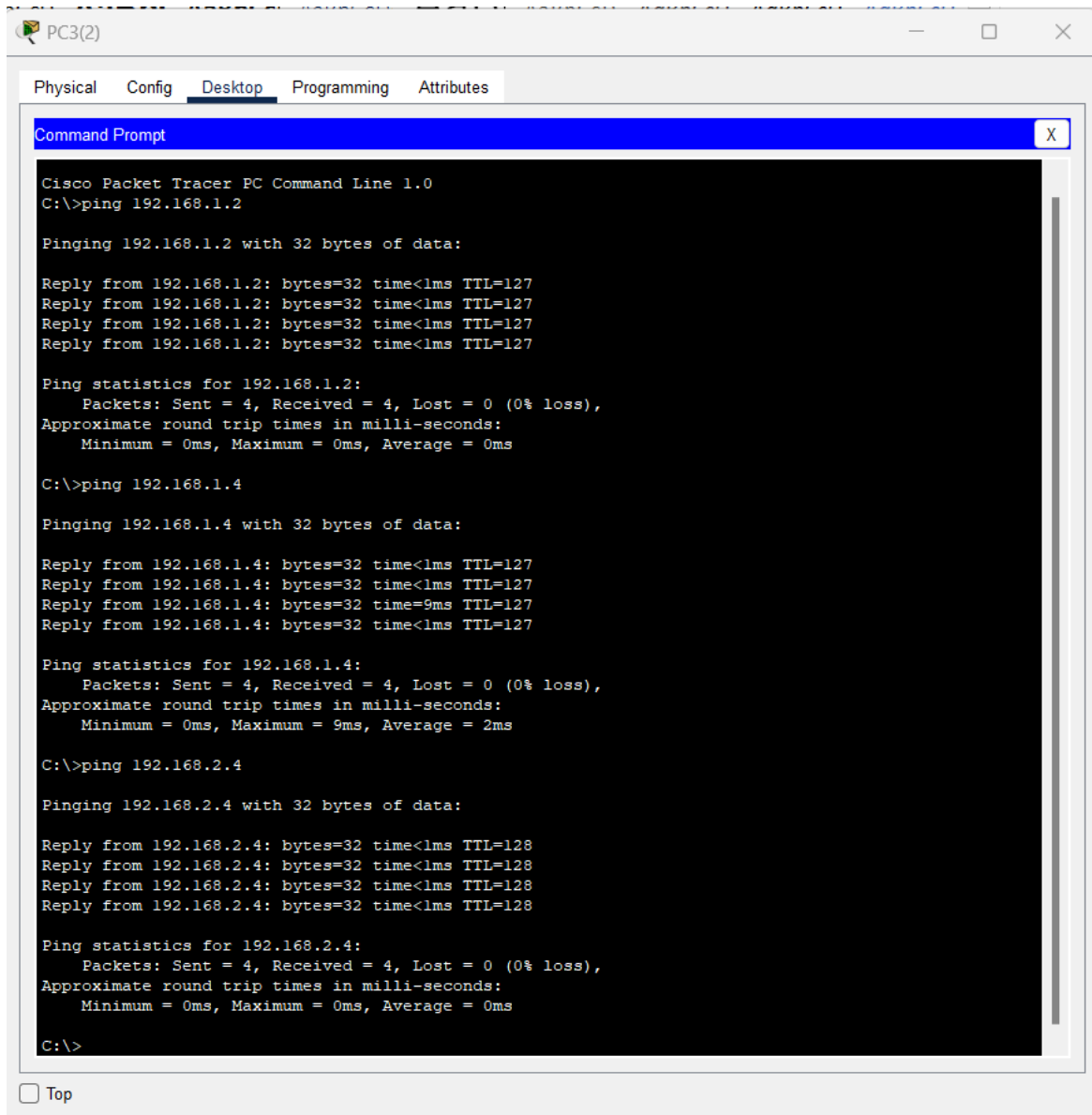
Pinging 192.168.2.4 with 32 bytes of data:

Reply from 192.168.2.4: bytes=32 time=9ms TTL=127
Reply from 192.168.2.4: bytes=32 time=1ms TTL=127
Reply from 192.168.2.4: bytes=32 time<lms TTL=127
Reply from 192.168.2.4: bytes=32 time<lms TTL=127

Ping statistics for 192.168.2.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>
```

PC11 (PC 3(2))



PC3(2)

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=127
Reply from 192.168.1.4: bytes=32 time<1ms TTL=127
Reply from 192.168.1.4: bytes=32 time=9ms TTL=127
Reply from 192.168.1.4: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>ping 192.168.2.4

Pinging 192.168.2.4 with 32 bytes of data:

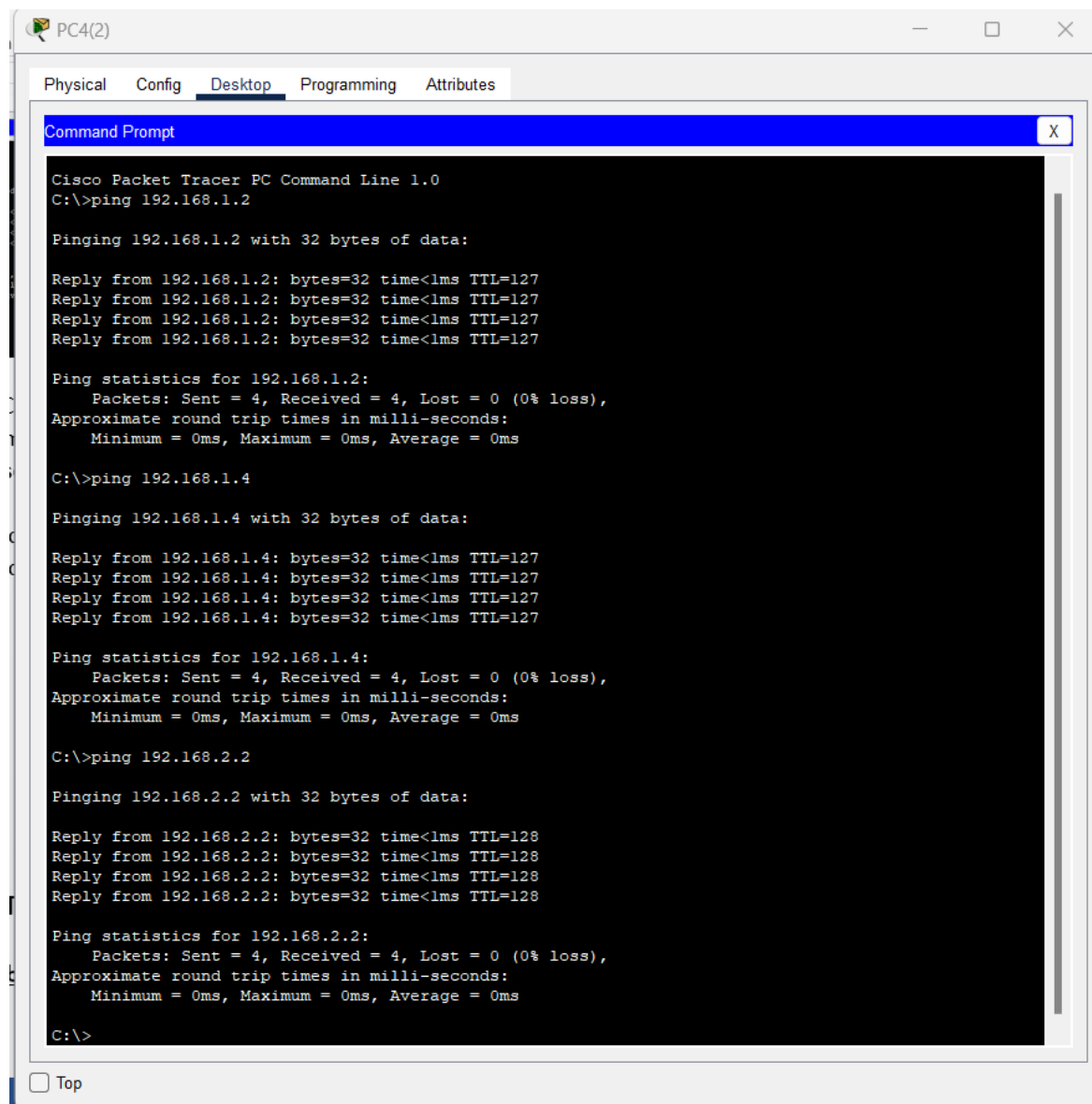
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

☐ Top

PC22 (PC4 (2))



```
PC4(2)
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=127
Reply from 192.168.1.4: bytes=32 time<1ms TTL=127
Reply from 192.168.1.4: bytes=32 time<1ms TTL=127
Reply from 192.168.1.4: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Part C: Validation

Step 1: Verify your configuration

a) Route 1# **show version**

What is Router model? Cisco 2621

What is Cisco IOS software version? IOS (tm) C2600 Software (C2600-I-M), Version 12.2(28)

What is IOS file name: flash:c2600-i-mz.122-28.bin

What is interface available in this router? 2 FastEthernet/IEEE 802.3 interface(s)

What is RAM size: 32K bytes

What is NVRAM size: 63488K bytes

b) Route 1# **show running-config**

What is purpose of this command?

Display configuration currently running in RAM

c) Route 1# **show ip interface brief**

List down all details of the interfaces with their status

```
Router1#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          192.168.1.254   YES manual up          up
FastEthernet0/1          192.168.2.254   YES manual up          up
```

d) Route 1# **show ip protocols – why no output displayed?**

The command is to display the parameters and current state of the active routing protocol process. At the moment no routing protocol is set up.

e) Route 1# **show ip route**

List two directly connected interface to this router.

```
Router1#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          192.168.1.254   YES manual up          up
FastEthernet0/1          192.168.2.254   YES manual up          up
Router1#sho ip protocols
Router1#show ip protocols
Router1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.1.0/24 is directly connected, FastEthernet0/0
C    192.168.2.0/24 is directly connected, FastEthernet0/1
```